

1J-4

Suppose a point P moves on the surface of a sphere with center at the origin. Let

$$\vec{OP} = \mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}.$$

- (a) Using the x, y, z coordinates, show that the velocity vector $\mathbf{v}$ is always perpendicular to $\mathbf{r}$.
- (b) Without coordinates (use the formula in 1J-3 which is valid also in space).
- (c) Prove the converse: If $\mathbf{r}$ and $\mathbf{v}$ are perpendicular, then the motion of P is on the surface of a sphere centered at the origin.

1J-5

- (a) Suppose a point moves with constant speed. Show that its velocity vector and acceleration vector are perpendicular. (Use the formula in 1J-3.)
- (b) Show the converse: If the velocity and acceleration vectors are perpendicular, the point P moves with constant speed.

1J-6

For the helical motion

$$\mathbf{r}(t) = a \cos t \mathbf{i} + a \sin t \mathbf{j} + bt \mathbf{k},$$

- (a) Calculate $\mathbf{v}$, $\mathbf{a}$, and $\mathbf{T} = \frac{d\mathbf{s}}{dt}$.
- (b) Show that $\mathbf{v}$ and $\mathbf{a}$ are perpendicular. Explain using 1J-5.

1J-7

- (a) Suppose you have a differentiable vector function $\mathbf{r}(t)$. How can you tell if the parameter t is the arclength s measured from some point in the direction of increasing t , without actually having to calculate s explicitly?

- (b) How should a be chosen so that t is the arclength if

$$\mathbf{r}(t) = (2a \cos t)\mathbf{i} + (2a \sin t)\mathbf{j}?$$

- (c) How should a and b be chosen so that t is the arclength in the helical motion described in Exercise 1J-6?

1J-8

- (a) Prove the formula

$$\frac{d}{dt}(u(t)\mathbf{r}(t)) = \frac{du}{dt}\mathbf{r}(t) + u(t)\frac{d\mathbf{r}}{dt}.$$

(You may assume the vectors are in the plane; calculate using the components.)

- (b) Let

$$\mathbf{r}(t) = e^t \cos t \mathbf{i} + e^t \sin t \mathbf{j},$$

the exponential spiral. Use part (a) to find the speed of this motion.

1J-9

A point P is moving in space with position vector

$$\mathbf{r} = \overrightarrow{OP} = 3 \cos t \mathbf{i} + 5 \sin t \mathbf{j} + 4 \cos t \mathbf{k}.$$

- (a) Show it moves on the surface of a sphere.
- (b) Show its speed is constant.
- (c) Show the acceleration is directed toward the origin.
- (d) Show it moves in a plane through the origin.
- (e) Describe the path of the point.

1J-10

The positive curvature k of the function $\mathbf{r}(t)$ is defined by

$$k = \left| \frac{d\mathbf{T}}{ds} \right|.$$

- (a) Show that the helix (Exercise 1J-6) has constant curvature. (It is not necessary to calculate s explicitly; calculate $\frac{d\mathbf{T}}{dt}$ instead and relate it to k using the chain rule.)
- (b) What is the curvature if the helix is reduced to a circle in the x - y plane?

1K Kepler's Second Law

1K-1

Prove the product rule for differentiating the dot product of two plane vectors. Do the calculation using an i, j coordinate system.

Let

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}.$$

1K-2

Let $\mathbf{g}(t)$ be a vector function. Prove by using components that

$$\frac{d\mathbf{g}}{dt} = 0 \quad \Longleftrightarrow \quad \mathbf{g}(t) = \mathbf{k},$$

where $\mathbf{k}$ is a constant vector.

1K-3

In our proof that Kepler's second law is equivalent to the force being central, we used the following steps to show the second law implies a central force.

Kepler's second law says the motion is in a plane and

$$2\frac{dA}{dt} = |\mathbf{r} \times \mathbf{v}|.$$

If

$$\frac{dA}{dt} = \text{constant},$$

then $\mathbf{r} \times \mathbf{v}$ is constant.

So,

$$0 = \frac{d}{dt}(\mathbf{r} \times \mathbf{v}) = \mathbf{v} \times \mathbf{v} + \mathbf{r} \times \mathbf{a} = \mathbf{r} \times \mathbf{a}.$$

This implies $\mathbf{a}$ and $\mathbf{r}$ are parallel, i.e., the force is central.
Reverse these steps to prove the converse.

Solutions

Vectors and Solutions

1A. Vectors

1A-1

$$\begin{aligned} \text{a) } |A| &= \sqrt{3}, & \text{dir } A &= \frac{A}{\sqrt{3}} \\ \text{b) } |B| &= 3, & \text{dir } A &= \frac{A}{3} \\ \text{c) } |A| &= 7, & \text{dir } A &= \frac{A}{7} \end{aligned}$$

1A-2

$$\frac{1}{25} + \frac{1}{25} + c^2 = 1$$

$$c = \pm \frac{\sqrt{23}}{5}$$

1A-3

a)

$$A = -\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}, \quad |A| = 3, \quad \text{dir } A = \frac{A}{3}$$

b)

$$A = 2\mathbf{i} + 4\mathbf{j} - 4\mathbf{k}$$

Let P be its tail and Q its head. Then

$$\overrightarrow{OQ} = \overrightarrow{OP} + A$$

$$= 4\mathbf{i} - 3\mathbf{k}$$

Therefore,

$$Q = (0, 4, 3)$$

1A-4

a)

$$\overrightarrow{OX} = \overrightarrow{OP} + \overrightarrow{PX}$$

$$= \overrightarrow{OP} + \frac{1}{2}(\overrightarrow{PO})$$

$$= \frac{1}{2}(\overrightarrow{OP} + \overrightarrow{OQ})$$

b)

$$\overrightarrow{OX} = s\overrightarrow{OP} + r\overrightarrow{OQ}$$

Use $1 - r = s$.

1A-5

$$A = \frac{3\sqrt{3}}{2}\mathbf{i} + \frac{3}{2}\mathbf{j}.$$

The condition is not redundant since there are two vectors of length 3 making an angle of 30° with $\mathbf{i}$.

1A-6

Wind

$$\mathbf{w} = 50(-\mathbf{i} - \mathbf{j})/\sqrt{2}, \quad \mathbf{v} + \mathbf{w} = 200\mathbf{j}.$$

$$\Rightarrow \mathbf{v} = \frac{50}{\sqrt{2}}\mathbf{i} + \left(200 + \frac{50}{\sqrt{2}}\right)\mathbf{j}.$$

1A-7

- a) $b\mathbf{i} - a\mathbf{j}$
- b) $-b\mathbf{i} + a\mathbf{j}$
- c)

$$\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2 = 1$$

$$\mathbf{j}' = -\frac{4}{5}\mathbf{i} + \frac{3}{5}\mathbf{j}.$$

1A-8

- a) This is elementary trigonometry.
- b)

$$\cos \alpha = \frac{a}{\sqrt{a^2 + b^2 + c^2}}, \quad \text{etc.}$$

$$\text{dir } A = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right).$$

c)

If t, u, v are direction cosines of some vector A , then

$$t\mathbf{i} + u\mathbf{j} + v\mathbf{k} = \text{dir } A$$

is a unit vector, so

$$t^2 + u^2 + v^2 = 1.$$

Conversely, if this relation holds, then

$$t\mathbf{i} + u\mathbf{j} + v\mathbf{k} = \mathbf{U}$$

is a unit vector, so $\text{dir } \mathbf{U} = \mathbf{U}$, and t, u, v are direction cosines of $\mathbf{U}$.

1A-9

Letting A and B be the two sides, the third is $B - A$.

The line joining the two midpoints is

$$\frac{1}{2}B - \frac{1}{2}A = \frac{1}{2}(B - A),$$

a vector parallel to the third side and half its length.